Doc R: Plaintext .m App File Format

What is doc requirements analysis and why do we do it?

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Summarize the Feature

The App Designer team will be introducing a new file format for apps. The current MLAPP file format has been meeting the needs of many users, but for users creating large apps and collaborating with other contributors over source control, the lack of editing flexibility, app running performance, and binary file format causes a lot of pain. The feature is a new, dual M+XML file format, where the M file contains user code, such as callbacks and properties, and the XML contains App Designer-managed code, such as the app layout and metadata. This two-file format provides better source control integration, better performance, and more flexibility in the app building process.

Gather Resources

How do I do this?

Resources	Type of resource?
RFA: File Format Preferences	RF spec
F: Multi-File M App	RF spec
RFA: matlab.apps.App	RF spec
F: Plain Text App Editor Integration and Open Behaviors	RF spec
FA: Metadata Serialization for Plain-Text File Format	RF spec
RFA: Appendix Debugging	RF spec
24a - Usability Test Plan for Plain Text Apps	Usability study notes
Plain Text Format WG	Meeting notes and timeline
http://p4db.mathworks.com/cgi-bin/depotTreeBrowser.cgi?FSPC=//doc/Bdoc/matlab/doc/src/nonrelease/matlab/26aPlaintextApps&HIDEDEL=YES	Location for PDF 1-pager for 26a prerelease plaintext customer engagement (see 26aPlaintextApps_book.xml)
Preliminary Interview Data - Plain-Text Study	UX Preliminary Interview notes (see Documentation column)

Understand User Roles and Goals

How do I do this?

User background (what does this user already know?)	PRISM user type	User role or persona	Goal
MATLAB user familiar with general App Designer workflows	Power User	User building an app for personal use	Use App Designer to build new apps and edit previously built apps
	Power User/ Developer	User collaborating on an app with others	Diff and merge changes to an app built in App Designer
	Power User/ Developer	User building an app as a tool for others	Share an app built in App Designer that all app users can run
	Power User/ Developer	User building an app for personal use	Make quick edits to app code or layout without launching MATLAB or App Designer

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Define Workflows for Goals

How do I do this?

Workflows...

Use App Designer to build new apps and edit previously built apps

Workflow Steps

1. Open new or existing app for editing

Question	Answer	Where this information will be covered
What file format is my app?	MLAPP by default	All existing doc - no changes.
How do I know whether I should switch my file format?	• If the MLAPP file format is working for you, there is no need to switch. • If you use your file with source control or want to edit your code in the MATLAB desktop or a different IDE, consider using the dual file format.	
How do I convert an existing app to use M+XML?	Save As or Save Copy	

	As	
How can I change the default file format?	MATLAB settings panel	

Share an app that all app users can run

Workflow Steps

1. Develop an app using your preferred file format.

Question	Answer	Where this information will be covered
I know I want to share my app broadly. Should that impact my choice of file format?	• M+XML apps will not run in release before R2026a. If you want to share with users on older releases, you will either need to save a copy of your app as an MLAPP file or develop	

	your app using the MLAPP format from the start.	
How do I choose my file format?	• Set your default format in MATLAB settings. • Open App Designer, click New Blank App	

2. When the app is ready to be shared, save a copy using the MLAPP file format

Question	Answer	Where this information will be covered
How do I save an app in a different file format?	Save As or Save Copy As options in the toolstrip	

3. Share the MLAPP file with users running MATLAB in previous releases

Question	Answer	Where this information will be covered

What will an app user in an older release see if I share an M+XML file instead of an MLAPP file?	They will get an error that the app can't run, and have an option to download a tool to convert the file.	

Workflow Context

Question	Answer	Where this information will be covered
What alternative approaches should users know about?	• Users can still develop their apps using the MLAPP file format to conserve compatibility	
What prerequisites do users need?		
How will users apply the results?	• App users will be able to run the app in older releases	
When should users not apply this workflow?	• When an app author will need to *develop* their app in previous releases, it might be easier to use the MLAPP format rather than keep track of two copies of the same app	
What measures of success should users look for?	• Being able to see inside their app file outside of	

	App Designer/in their source control system	
What default behavior in the workflow can users modify?	N/A	
What limitations should users consider?	• Editing the file outside of App Designer can lead to the app becoming corrupted/unable to open or run	

Diff and merge changes to an app

Workflow Steps

1. Create an app that's compatible with source control.

Question	Answer	Where this information will be covered
What App Designer apps are compatible with source control?	Apps using the M+XML file format.	
What is the M+XML file format and why would I want to use it?	A new dual-file plaintext file format for apps created in App Designer, that's easier to diff and merge using source control.	
How do I create a new app using the M+XML file format?	Update your default file format in MATLAB serttings. Then, open App Designer and click New Blank App	
How do I convert an existing app to the M+XML file format?	Open your app in App Designer. Use Save As or Save Copy As in the toolstrip to save a copy of your app in the new format.	
How can I confirm that my app is using the	• File extension of app in App Designer is .m	

M+XML file format?	• You have two files associated with the app: *appname*.m and *appname*.xml. • In Code View, only your code is visible (not the App Designer generated code) • You can open both files in any editor and view their contents	

2. View changes to the app that a collaborator has made.

Question	Answer	Where this information will be covered
How do I view changes to my app file in source control?	Using the same workflow as diffing/merging any other plaintext files in source control.	

What do the two files consist of?	• M file:
	  ◦ App callbacks and properties, stored as MATLAB code
	• XML file:
	  ◦ App components and layout
	  ◦ App metadata, such as author and version
	  ◦ Internal data managed by App Designer	

3. Decide whether to accept or reject the changes to the app

Question	Answer	Where this information will be covered
How do I determine if a change will break the app?	Run the updated file in MATLAB or visually identify if the changes make sense.	

What part of the app (component, callback code, etc.) does the change relate to?	• In the M file: callbacks and app behavior • In the XML file: app component creation and configuration and app layout	

4. Merge some of the changes into the app

Question	Answer	Where this information will be covered
Are there parts of the file that I shouldn't merge?	• The XML appendix can be fragile, so if you're merging code there, it's a good practice to verify that the app still loads runs after the merge. • Avoid editing content in the XML file that represents the internal data managed by App Designer.	

5. Verify that the app runs and behaves as expected after the merge

Question	Answer	Where this information will be covered
How do I run the app?	From MATLAB (command line or from the Editor with the M file open) or App Designer	
What can I do if the app doesn't run or behave as expected?	Revert the merge, or troubleshoot by editing the app code by hand.	

Workflow Context

Question	Answer	Where this information will be covered

What alternative approaches should users know about?	• Users can still use the MLAPP file format and the built-in MATLAB diff/merge too.	
What prerequisites do users need?	• Users need an existing app in a source control system	
How will users apply the results?	• Diff/merge app files in the same way they diff/merge any other code file	
When should users not apply this workflow?	• When the user needs their app to run in previous versions of MATLAB	
What measures of success should users look for?	• Being able to see inside their app file outside of App Designer/in their source control system	
What default behavior in the workflow can users modify?	N/A	
What limitations should users consider?	• Editing the file outside of App Designer can lead to the app becoming corrupted/unable to open or run	

Make quick edits to app code or layout without launching App Designer

Workflow Steps

1. Open the M and/or XML file outside of App Designer

Question	Answer	Where this information will be covered
What programs can I open the file in?	Any. The M file is all MATLAB code, so the best experience will be to open it in MATLAB.	

How do I interpret what I'm seeing?	• The M file consists of all of the app logic that you have written • The XML file consists of configuration data that App Designer manages, such as component creation and layout	

2. Edit the callback code.

Question	Answer	Where this information will be covered
How do I edit my callback code?	Using your normal MATLAB code editing practices.	
What sorts of changes are safe?	Really any. This is just M code, you can edit it like any other M code.	

3. Make small tweaks to property values

Question	Answer	Where this information will be covered

How do a change a property value?	Find the element that represents the property on the component and update the value.	
What sorts of changes are safe?	• Editing a property	
What sorts of changes should I avoid making outside of App Designer?	• Editing App Designer-managed data • Large layout changes (deleting or adding components, etc) – edit at your own risk.	

4. Confirm that the app opens and runs as expected

Question	Answer	Where this information will be covered
How do I run the app after making changes outside of App Designer?	Same as any app: open the file in AD and run it there, or run it from MATLAB at the Command Window or	

	from the CFB	
What happens if I made edits that aren't compatible with the file structure?	• On run: You'll see an error • On load: Either App Designer will try to recover and update the code, OR the app will fail to load with an error	
How do I recover from a file that App Designer can't load?	• Make sure the code in the XML file matches the format that App Designer expects (*todo: will we have an XSD for customers to validate against?)*	

Documenting the Appendix

Click here to expand...

Reviewer Comments for IDR

- Expect file format to be documented - explain concept of canvasGroup, for instance (Michelle)
- Need to be told what parts of the file should I be editing, not editing. Esp if it can break my app loading back into AD (Michelle)
- Need to be taught to ignore the file hash (Michelle)
- Can users do the merge if it is non trivial? How does the users reason on it when there are many changes. (John G)
- At some point if I wanted to see what was fully going on, I might want to see all of the code. For example, the information in the Inspector... where is Balloons and Fish. Its not here anywhere in this file. At some point I would start wondering, but hadn't yet. (Vadim)
- Should I be modifying / allowed to modify [file hash, UUID, other metadata ...]? (Multiple reviewers)

Reviewers also expressed that the XML content was easy to read/reason on just by looking at.

When will users interact with the appendix?

Task	Information needed
Interpret a diff to decide whether to accept changes to an app	• Understanding of how XML elements and attributes map to App Designer concepts ◦ *The ComponentName attribute changed. What will this look like if I open or run the app?* • Understanding whether the changes result in an artifact that "works" (successfully runs/loads in App Designer) ◦ *The diff shows that a callback function was deleted. Will this result in errors if I try to run the*

	app?
Perform a merge of an app file	In addition to the info needed to interpret a diff... • Know if the resulting XML is well-formed ◦ *I merged a lot of different elements and attributes. Is the end result something that App Designer will be able to open?* • Understanding of which parts of the file can be edited and any risks associated with manual editing ◦ *There's a merge conflict in the file hash. Which edit should I accept?*
Check that an action in the design environment had the intended effect	• Information about how to view appendix ◦ *Where is the code that controls how my app is laid out?* • Understanding of how XML elements and attributes map to App Designer concepts ◦ *I changed the icon of my button. Where can I see this change reflected in the code?*
Make quick edits to component configuration	• Understanding of which parts of the file can be edited and any risks associated with manual editing ◦ *I want to quickly change the color of a panel. Can make that change in VS Code? Is it possible I can corrupt my app if I enter the color format incorrectly?*
Recover from an issue in the appendix code	• Know if the resulting XML is well-formed ◦ *What part of the document is causing the issue?* • Know the format that App Designer needs to read the data ◦ *If I delete this element entirely, what will happen when I try to open the app in App Designer?*

Design Doc Plan

How do I do this?

Release Notes

Products	Compatibility note?	Feature highlight?	Title
MATLAB	No	TBD, but likely	App Designer: Use new plaintext MAPP file format for better source control integration and performance

Reference

New or update?	Type (function/class/object...)	Title	Purpose	Required for 26b prerelease?
Update	Tool	App Designer	Update file format references and screenshots	N
Update	settings API	matlab.appdesigner settings	New settings group under appdesigner (see A section of this RFA spec: RFA: File Format Preferences)	N
Update	Settings	App Designer Settings	Include file format preferences	N
~~New~~	~~Abstract class~~	~~matlab.apps.App~~	• ~~Provide a breadcrumb that advanced/curious users can use to learn more about the base class for plaintext apps~~	~~Y~~

			• ~~Document the lifecycle/execution order of an app, including:~~
			  ◦ ~~When the XML file is accessed.~~
			  ◦ ~~Which class accesses it.~~
			  ◦ ~~How the data flows from XML to UI Components.~~
			• ~~Make it clear that this is a class that App Designer uses to create apps, and isn't meant to be something that a user subclasses~~
			
			~~See~~ Discussion: Reference Pages for AppBase and App Classes ~~for proposal, mockups, and discussion notes.~~	

~~New~~	~~Abstract class~~	~~matlab.apps.AppBase~~	• ~~Provide a breadcrumb that advanced/curious users can use to learn more about the base class for plaintext apps~~ • ~~Provide a breadcrumb for some of the back-box methods that users have asked about (e.g. g2732146)~~ • ~~Make it clear that this is a class that App Designer uses to create apps, and isn't meant to be something that a user subclasses~~ ~~See~~ Discussion: Reference Pages for AppBase and App Classes ~~for proposal, mockups, and discussion notes.~~	~~Y~~

Topics

New or update?	Title	Overview of Content	Required by 26b prerelease?

New	App Designer File Formats	• Overview of M+XML and MLAPP files
		  ◦ Breakdown of how user code vs. generated code is represented
		  ◦ Decision-making support about when to use each, including what isn't supported (e.g. UAC? Simulink apps?) → represented as a table for quick decision making
		• How to modify your default file format
		  ◦ Preferences
		• How to convert apps between formats
		  ◦ Save As
		  ◦ Convert M+XML to MLAPP for backwards compatibility
			Y

New	MATLAB App File Format (.m + .xml)	• *Similar to live code page:* Live Code File Format (.m) - MATLAB & Simulink
		• *Proposing after 4/2/26 xml schema overview meeting:* Doc R: Plaintext .m App File Format | 4/2/2026
		
		• M: How code is represented
		  ◦ User code in M file
		  ◦ Layout and data in XML file
		  ▪ Component and properties as elements
		  ▪ Design-time data as attributes
		  ▪ Metadata
			Y?

		◦ What is required to run/what will be automatically assumed default	
Update	Compatibility Between Different Releases of App Designer	• Compatibility of different App Designer files ◦ Running/editing/sharing MLAPP files between releases ◦ Running/editing/sharing M+XML files between releases ◦ Convert M+XML to MLAPP for backwards compatibility	Y
Update	Compare and Merge Apps	• KEEP information on merging MLAPP file • ADD a section to describe merging M+XML files ◦ *What the diff/merge workflow looks like in the MATLAB comparison tool (workflow to be designed)* ◦ High-level overview of how data is represented in the appendix, with a link to the App Designer File Formats topic for more info ◦ Best practices for merging ▪ General "edit at your own risk" note ▪ Warning not to edit data managed by App Designer, and info about how to identify that data in the appendix	Y
Update	Manage Code in App Designer Code View	• Add callout about M+XML file format, with a link to more info about the different file formats.	N
Update	Startup Tasks and Input Arguments in App Designer	• Add information about when the StartupFcn executes w/r/t the rest of app startup, since this isn't self-documenting in the M+XML file.	N
New?	*TBD – any troubleshooting info needed? Potentially a new debugging topic that covers both files? Probably*		N

	not needed, but something to keep on the radar		

New?	Binary App File Format	• To parallel page on plain text app format
		• Live Code File Format (.mlx) - MATLAB & Simulink
		• MLAPP: How code is represented
		  ◦ All code in MLAPP file
		  ◦ User code writeable, layout and data read-only
			N

Example Apps

Most existing MLAPP examples will not be converted for R2026a, since MLAPP is still the default file format and the task the examples are demonstrating aren't dependent on the file format.

The examples that *will* be converted to M+XML are:

Example	Rationale for Conversion
Organize App Data Using MATLAB Classes	This example is meant for a similar audience as the plaintext file format: Customers will large apps who might be collaborating on them with others.

Notes

4/2/2026

- Plain Text App XML Schema Overview (26b)
- XML schema info needs to be accessible to users in the first release (see first comment)
- PROPOSE: A topic page for the plain text file format

AD Check-in, 3/20/2026

- Discussion: App Designer File Formats topic pages

Compare and Merge Apps – Updates/Questions

- Cannot compare binary and plain text app code
- Is **Compare To** the recommended way to compare .m files for plain text apps? Can .xml files b compared/is there any reason to do so?
- Is App Designer-managed code identified in the same way as .mlapp apps (gray background)?

AD Check-in, 02/10/2026

We decided not to document the base classes matlab.apps.App and matlab.apps.AppBase beca they are not designed in a way to be used outside of App Designer (see Jyoti's comments above) This is in contrast to the documented base classes for authoring components and charts:

- https://www.mathworks.com/help/matlab/ref/matlab.ui.componentcontainer.componentcontai class.html
- https://www.mathworks.com/help/matlab/ref/matlab.graphics.chartcontainer.chartcontainer-class.html

There should also be further discussions on how much of the xml structure to document, as this design may evolve in future releases.

Next Steps:

- [ ] Review gecks involving base classes and execution order. What questions are they asking and can they be addressed elsewher the doc?

AD Check-in, 7/01/2025

Without the generated code of the MLAPP file, IDR reviewers gave the feedback that how the ap executes (e.g. when and how components are created using the XML data) feels like magic. To mitigate this, the team decided to add additional comments in the M file, and also to document th execution order of the plaintext app file somewhere in the doc. For more info, see F: XML Proces Transparency.

In follow-up conversations with the team, we decided to just make it more clear on the topic that talks about the StartupFcn what the execution order is (since with the plaintext format there isn't self-documenting generated code). If we need more, it can probably be in the matlab.apps.App reference page.

AD Check-in, 4/22/2025

As of today, if users write any custom code in their M file that doesn't conform to the way AD expects the code to look (e.g. different order of properties/callbacks, added local function...), AD delete that code when the app is loaded. There are discussions about addressing this before shipping plain-text, but if that doesn't converge, we will definitely need something in the doc to w users of pitfalls.

AD Check-in, 1/28/2024

Alex's internal API for determining plaintext or not proposes not explicitly supporting indirect inheritance for matlab.apps.App (both creating and running). We might want to call this out somewhere in the doc when we talk about programmatically working with app files – likely on the matlab.apps.App reference page.

Re: RF: API to Identify if M File is Plain Text App

Doc Design Review, 4/2/2024

Agenda:

- Review Understand User Roles and Goals, and Design Doc Plan sections (*skip the workflow sections)*
- Discuss doc plan. A couple topics for conversation:
  - Level of support for editing code outside of App Designer (user goal that we're explicitly targeting? or side effect of merge workflows?)
  - Implications of documenting appendix (what's an incompatibility? possibility of users codin against anything we document?)
  - Comfort/priority of updating doc examples to use MAPP (maybe bring this as a future topic, not sure we're ready to answer this at this point in the design)

Leave Feedback for Doc R Templates

Related Information

- Research Doc Requirements
- Research Doc Requirements for a Feature
- Sample Feature Doc Requirements

- Customer-Focused Questions